Supplementary material

How Much of a Time Series Survives Its Image? A Cross-Domain Comparative Analysis of the Reconstruction Capacity of Image-Based Representations

S1. Classifier settings

All classifiers are trained on original windows, standardised per channel with the statistics of the training fold, and evaluated on the original and on every reconstruction of the same held-out windows. KNN ($k = 5$), an RBF SVM ($C = 1$, Platt probabilities, one-vs-rest for the multilabel PTB-XL) and a random forest (300 trees) take the flattened window. The LSTM and GRU (64 and 32 units, dropout 0.2) and a one-dimensional CNN (convolutions of 32 and 64 filters of width 5, max pooling, global average pooling and a 64-unit dense layer) take the (T, C) window and are trained with Adam, batch size 64 and at most 20 epochs, with early stopping (patience 4) on the last 10% of the training fold. Metrics are computed per fold and averaged over the three folds; the equivalence intervals resample the groups of the split (subjects, patients, events, stations or recordings).

S2. Fidelity at the other window lengths

Tables S1 and S2 repeat Table 3 of the main text for $T = 64$ and $T = 256$. EMG at $T = 256$ has fewer than 200 windows and is not reported.

Table S1: Mean 8-bit NMAE of every encoding–decoder combination at $T = 64$. Oracle: geometric decoder on the unsigned RP with the reflection picked using the original series.

	Hirata		Sipers		Geometric (ours)						
Dataset	Uns.	Sig.	Uns.	Sig.	Uns.	Oracle	Sig.	GASF	GADF	MTF	XIRP
WISDM	0.2647	0.1171	0.2084	0.0042	0.2039	0.0005	0.0012	0.0016	0.0011	0.2930	0.0003
PAMAP2	0.2630	0.1153	0.2108	0.0049	0.2103	0.0008	0.0017	0.0017	0.0012	0.2889	0.0003
ESC-50	0.2463	0.0956	0.1997	0.0056	0.2247	0.0007	0.0016	0.0017	0.0010	0.2937	0.0003
PTB-XL	0.3530	0.1916	0.2991	0.0060	0.2910	0.0010	0.0022	0.0033	0.0019	0.3434	0.0003
Swarm	0.2710	0.0293	0.2613	0.0028	0.2630	0.0129	0.0144	0.0025	0.0013	0.2942	0.0003
STEAD	0.2361	0.1001	0.2052	0.0055	0.2030	0.0006	0.0014	0.0016	0.0010	0.2733	0.0003
EMG	0.2300	0.1365	0.1869	0.0039	0.1947	0.0004	0.0010	0.0014	0.0020	0.2982	0.0006

Table S2: As Table S1, at $T = 256$.

	Hirata		Sipers		Geometric (ours)						
Dataset	Uns.	Sig.	Uns.	Sig.	Uns.	Oracle	Sig.	GASF	GADF	MTF	XIRP
WISDM	0.2228	0.0971	0.1743	0.0046	0.1682	0.0004	0.0010	0.0014	0.0015	0.2700	0.0002
PAMAP2	0.2084	0.0960	0.1829	0.0053	0.1736	0.0006	0.0015	0.0015	0.0015	0.2740	0.0002
ESC-50	0.1859	0.0752	0.1814	0.0058	0.1690	0.0007	0.0017	0.0015	0.0012	0.2753	0.0001
PTB-XL	0.3066	0.1600	0.2588	0.0051	0.2716	0.0006	0.0012	0.0025	0.0026	0.3291	0.0002
Swarm	0.2950	0.0339	0.3111	0.0155	0.2573	0.0108	0.0174	0.0027	0.0009	0.3066	0.0001
STEAD	0.1723	0.0789	0.1581	0.0047	0.1552	0.0005	0.0012	0.0014	0.0011	0.2592	0.0001
EMG	–	–	–	–	–	–	–	–	–	–	–

S3. Recurrence-rate sensitivity of Hirata's reconstruction

Table S3 gives the 8-bit NMAE of Hirata's reconstruction against the recurrence rate used to threshold the image ($T = 128$, 100 windows per dataset, every channel). The rate of 0.75 used in the main text was fixed in advance; choosing the best rate per dataset with knowledge of the result lowers the error by at most 0.025 on the unsigned and 0.010 on the signed RP.

Table S3: Hirata's reconstruction: 8-bit NMAE against the recurrence rate.

Dataset	Encoding	0.02	0.05	0.1	0.2	0.3	0.5	0.7	0.75	Best
WISDM	unsigned	0.239	0.250	0.248	0.245	0.239	0.240	0.237	0.231	0.75
	signed	0.225	0.196	0.171	0.136	0.120	0.103	0.104	0.103	0.75
PAMAP2	unsigned	0.240	0.255	0.252	0.238	0.236	0.231	0.233	0.238	0.5
	signed	0.222	0.194	0.161	0.124	0.105	0.098	0.102	0.107	0.5
ESC-50	unsigned	0.237	0.253	0.242	0.251	0.236	0.197	0.215	0.215	0.5
	signed	0.217	0.156	0.115	0.088	0.078	0.076	0.079	0.081	0.5
PTB-XL	unsigned	0.307	0.311	0.320	0.323	0.326	0.330	0.342	0.328	0.02
	signed	0.290	0.287	0.285	0.277	0.272	0.252	0.213	0.193	0.75
Swarm	unsigned	0.273	0.265	0.287	0.291	0.293	0.296	0.293	0.290	0.05
	signed	0.064	0.033	0.021	0.020	0.024	0.028	0.028	0.028	0.2
STEAD	unsigned	0.239	0.251	0.235	0.222	0.224	0.219	0.221	0.212	0.75
	signed	0.216	0.162	0.124	0.096	0.085	0.079	0.084	0.085	0.5
EMG	unsigned	0.214	0.234	0.234	0.220	0.215	0.201	0.200	0.203	0.7
	signed	0.199	0.192	0.163	0.138	0.126	0.116	0.119	0.121	0.5

S4. Sampling rate at a fixed length and at a fixed duration

Table S4 separates the sampling rate from the window duration on PTB-XL. With 128 samples, a higher rate also means a shorter window; with a fixed duration of 1.28 s, the number of samples grows with the rate. In

both settings the conditioning σ_2/σ_1 falls and the error of the signed RP rises with the rate, while GASF, GADF and XIRP do not follow that trend. The "native" rows use the 100 Hz files distributed with the dataset instead of resampling the 500 Hz records.

Table S4: PTB-XL, anti-aliased resampling from 500 Hz: median σ_2/σ_1 of the RP states, share of series below 0.1, and mean 8-bit NMAE.

Window	Rate	Seconds	σ_2/σ_1	Share < 0.1	Signed RP	Sipers / sig.	GASF	GADF	XIRP
128 samples	500 Hz	0.26	0.08	67%	0.0076	0.0078	0.0032	0.0018	0.0005
	250 Hz	0.51	0.16	23%	0.0041	0.0074	0.0032	0.0019	0.0002
	100 Hz	1.28	0.37	0%	0.0015	0.0055	0.0029	0.0022	0.0002
	100 (native) Hz	1.28	0.37	0%	0.0015	0.0054	0.0028	0.0022	0.0002
	50 Hz	2.56	0.61	0%	0.0009	0.0040	0.0021	0.0019	0.0002
1.28 s	500 Hz	1.28	0.08	76%	0.0079	0.0061	0.0028	0.0032	0.0002
	250 Hz	1.28	0.16	9%	0.0026	0.0064	0.0028	0.0028	0.0002
	100 Hz	1.28	0.37	0%	0.0015	0.0055	0.0029	0.0022	0.0002
	100 (native) Hz	1.28	0.37	0%	0.0015	0.0054	0.0028	0.0022	0.0002
	50 Hz	1.28	0.64	0%	0.0012	0.0037	0.0024	0.0016	0.0003

S5. Sipers et al. and the geometric decoder on the same image

The reconstruction of Sipers et al. assumes a periodic signal and a circular delay embedding, so in the main benchmark it decodes the circular variant of the RP, with the wrap state (x_T, x_1) added, while the geometric decoder decodes the benchmark's non-circular RP. Table S5 removes that difference: both decoders receive the identical circular image ($T = 128$, 100 windows per dataset, every channel, 8-bit), the same pixel matrix and the same per-channel calibration, and on the signed RP both choose the reflection from the image alone with the same criterion, so any gap is due to the decoder. The circular RP is built on the states (x_i, x_{i+1}), $i = 1, \ldots, T$, with $x_{T+1} = x_1$: it is the benchmark RP with one added state, the wrap state (x_T, x_1), and the same 8-bit maps.

Sipers' method represents the window by its periodic extension, which a finite sequence always admits, so its hypotheses hold for the circular embedding of any window; the cost is the wrap state, which joins the last sample to the first and is not part of the recorded dynamics. With a node threshold of zero, the implementation is exact in floating point (median error 0; maximum 2e-13 on WISDM, Swarm and STEAD, and 2.7e-03 for the worst of 1 200 PTB-XL series, where near-zero Fourier coefficients enter the division of Eq. (15) of

Sipers et al.); the error of the threshold used in the main text is therefore truncation, not a property of the method. The node threshold is a share of the largest Fourier power estimated from the image; it does not use the original series. It was chosen as the value that minimises Sipers' own 8-bit error in this sweep, which favours the method in the comparison. With the threshold of 0.01 used by its authors, the error is 0.009 to 0.044 even in floating point, because our windows are broadband. The last columns sweep the node threshold of the Sipers decoder, the share of the largest Fourier power below which a frequency is discarded; Sipers et al. used 0.01 without optimising it, and the main text uses 10^{-4}, the best value at 8 bits in this sweep.

Table S5: Mean 8-bit NMAE on the same circular RP. Unsigned: reflection chosen by an oracle (geometry alone). Signed: reflection chosen from the image by the sign, as in the main text.

	Unsigned, oracle sign		Signed		Sipers, unsigned, oracle sign, node threshold				
Dataset	Sipers	Geometric	Sipers	Geometric	0.01	0.001	0.0001	1e-06	0
WISDM	0.0026	0.0004	0.0042	0.0010	0.0440	0.0092	0.0026	0.0031	0.0034
PAMAP2	0.0033	0.0006	0.0050	0.0014	0.0296	0.0082	0.0033	0.0042	0.0043
ESC-50	0.0041	0.0006	0.0054	0.0013	0.0343	0.0099	0.0041	0.0049	0.0049
PTB-XL	0.0035	0.0007	0.0054	0.0014	0.0216	0.0052	0.0035	0.0050	0.0054
Swarm	0.0019	0.0010	0.0046	0.0025	0.0410	0.0379	0.0019	0.0020	0.0020
STEAD	0.0040	0.0005	0.0054	0.0012	0.0271	0.0078	0.0040	0.0055	0.0061
EMG	0.0015	0.0003	0.0039	0.0008	0.0093	0.0016	0.0015	0.0017	0.0017

S6. Computational cost

Table S6 gives the single-thread cost of every combination against the window length.

S7. Spectral and correlational fidelity

Table S7 separates the two reasons for a large pointwise error discussed in the main text: decoders that return the reflection keep the spectrum and the energy, while those that lose the waveform distort them.

Table S6: Median single-thread time (ms) to encode, store and decode one channel (WISDM, 20 series per length), log–log slope of time against T, and peak traced memory at $T = 1024$.

Decoder	$T = 64$	$T = 256$	$T = 1024$	Exponent	MB at 1024
Hirata / uns.	2.0	58.5	2871.2	2.64	87
Hirata / sig.	2.3	71.8	2880.1	2.59	87
Sipers / uns.	1.1	27.1	1172.7	2.47	112
Sipers / sig.	1.5	29.9	1193.1	2.45	112
Ours / uns.	0.6	13.1	398.2	2.35	72
Ours / sig.	1.3	25.1	552.0	2.22	72
GASF	0.1	1.0	12.0	1.85	32
GADF	0.1	1.0	13.0	1.72	32
MTF	0.5	1.5	12.8	1.22	24
XIRP	0.1	0.6	12.8	2.03	32

Table S7: Spectral and correlational fidelity after 8-bit storage: range of the dataset means over the seven datasets at $T = 128$. Pearson: correlation with the original; PSD shape: L_1 distance between unit-mass Welch spectra; Energy: relative error of total power.

Decoder	Pearson	PSD shape	Energy
Hirata / uns.	-0.023–0.013	0.0550–0.6117	0.224–0.766
Hirata / sig.	0.556–0.982	0.0562–0.6053	0.225–0.730
Sipers / uns.	-0.075–0.038	0.0021–0.0125	0.005–0.012
Sipers / sig.	0.999–1.000	0.0063–0.0260	0.010–0.020
Ours / uns.	-0.036–0.068	0.0007–0.0151	0.001–0.036
Ours / sig.	0.999–1.000	0.0037–0.0187	0.002–0.055
GASF	0.999–1.000	0.0041–0.0171	0.005–0.012
GADF	1.000–1.000	0.0005–0.0048	0.001–0.002
MTF	-0.021–0.009	0.3279–0.7871	1.112–3.154
XIRP	1.000–1.000	0.0003–0.0017	0.000–0.001